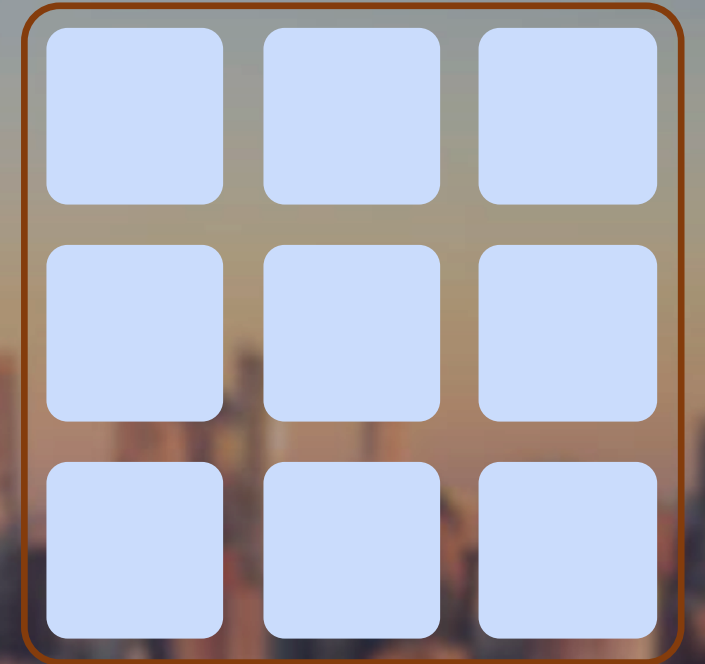


URBAN DATA ANALYSIS — FINAL PRESENTATION

Could Manhattan Have Superblocks?

*A data-driven search for Barcelona-style superblock candidate sites in the
Manhattan street grid*



A Barcelona Observation

- In the Eixample, wide chamfered intersections and traffic-calmed interior streets made the streetscape work for pedestrians and drivers alike — daily life fit within a short walk.
- The Manhattan grid looks similar on a map, but feels very different on the ground. Is that gap locked into the grid's form — or could Manhattan's streets be reallocated the way Barcelona's have been?
- Barcelona's superilles (superblocks) group 3×3 blocks and remove through-traffic from interior streets — because streets are the largest publicly-owned land reserve in a built-out city.
- Manhattan faces the same condition: buildings are fixed, but roughly a quarter of its land area is street — and it is publicly owned.



BARCELONA BLOCK

113 m × 113 m



MANHATTAN BLOCK

80 m × up to 280 m

Similar footprint, very different geometry — and, as this project asks, very different street behavior.

Where in the Manhattan street grid could Barcelona-style superblocks plausibly be implemented?

Which areas satisfy the network, traffic, and transit conditions that made superblocks feasible in Barcelona — and how much of Manhattan is structurally excluded?

Three Reasons This Question Is Worth Asking Now

1

Policy relevance and timing

NYC is already reallocating street space — Open Streets, pedestrian plazas, congestion pricing since 2025. A feasibility map speaks directly into that conversation.

2

A genuine open question

Manhattan's grid only looks like Barcelona's on the surface: elongated blocks, heavy avenue traffic and bus routes, extreme density. The answer isn't known in advance.

3

Builds on published research

Eggimann (2022, Nature Sustainability) screened 18 world cities using OpenStreetMap alone. This project localizes that method for Manhattan with real city-specific data.

DATASET

Six Real Datasets, One Screening Logic

All pulled live from public APIs — no synthetic data in the final results.

Dataset	Source	What we got
Street network	OSM / osmnx	4,634 nodes, 9,916 edges — base graph
NYC Street Centerline	NYC DCP (inkn-q76z)	30,341 segments; official street data
Traffic volume counts	NYC DOT (7ym2-wayt)	1,204 stations → 142 edges matched
Bus routes (GTFS)	MTA static feed	151 Manhattan bus shapes
Truck route network	NYC DOT (jjja-shxy)	32,939 segments
Population by block group	US Census ACS	1,292 block groups, 1.45M people

A Four-Stage Screening Pipeline



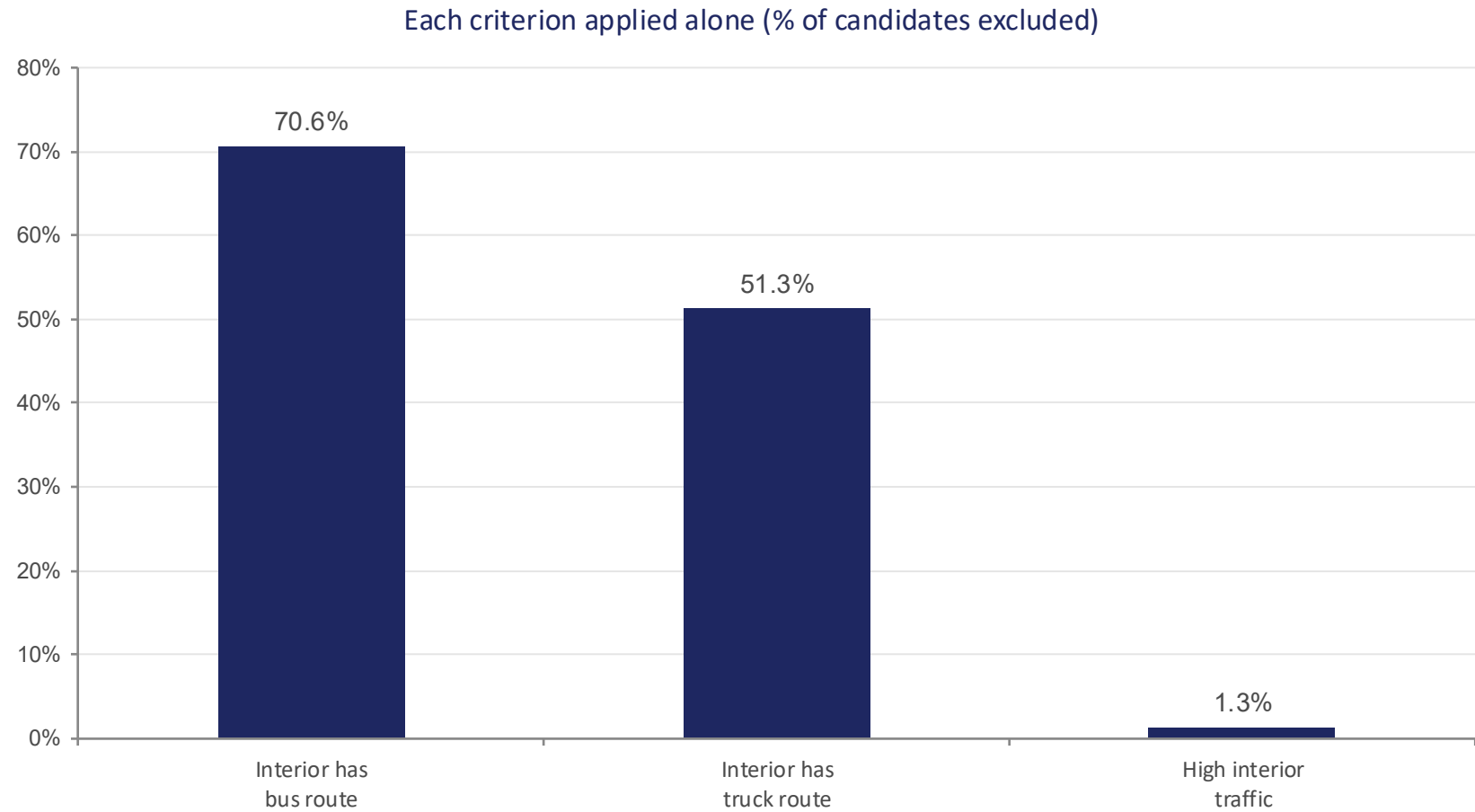
Candidate unit: a sliding window bounded by an avenue pair and a run of cross streets — comparable in scale to Barcelona's 3×3 block, but elongated like Manhattan's real grid.

RESULTS

Bus and Truck Routes — Not Traffic — Rule Out Manhattan

23.8%

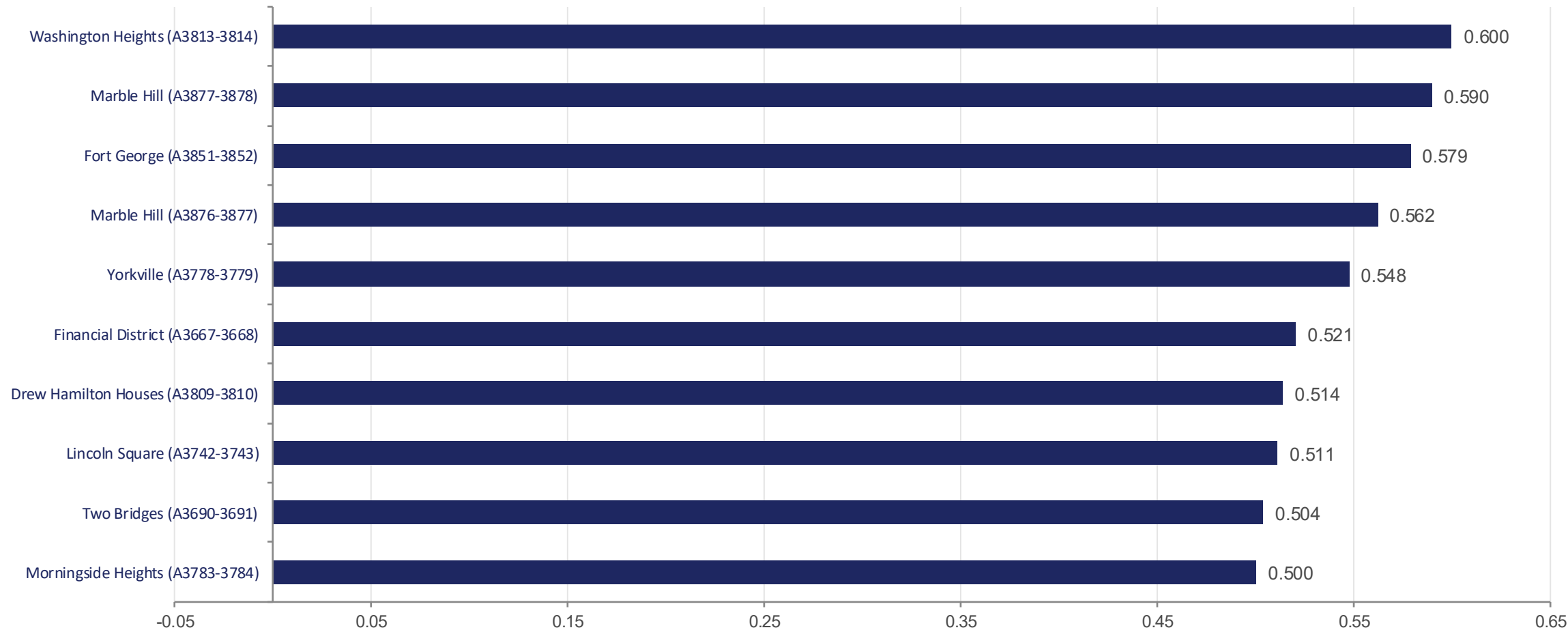
of candidates survive
1,483 of 6,241 units



Cumulative: 6,241 → 1,836 (after bus) → 1,484 (after truck) → 1,483 (traffic removes 1 more — most interior volumes were imputed near the exclusion threshold)

RESULTS

A Small Set of Strong Candidates — Not One Outlier



RESULTS

Where Are the Top Candidates, Actually?

unit_id is a grid index, not an address — reverse-geocoded here to real Manhattan places.

Rank	Neighborhood	Score (unit_id)
1	Washington Heights	0.600 (A3813-3814)
2	Marble Hill	0.590 (A3877-3878)
3	Fort George	0.579 (A3851-3852)
4	Marble Hill	0.562 (A3876-3877)
5	Yorkville	0.548 (A3778-3779)
6	Financial District	0.521 (A3667-3668)

RESULTS

The Ranking Holds Up Across Different Weightings

Spearman rank correlation between suitability rankings under three weighting schemes

0.99

Baseline
vs.
Traffic-averse

0.92

Baseline
vs.
Population-first

0.91

Traffic-averse
vs.
Population-first

All three comparisons are above 0.9 — the same neighborhoods surface as top candidates regardless of whether the model leans toward penalizing traffic or rewarding population served.

RESULTS

Closing the Top Candidate: The Grid Absorbs It

0.0%

change in avg. trip length
citywide test AND local test

57,014 → 58,258

perimeter capacity, before →
upper bound if 100% shifts (+2.2%)

Why 0.0% is a real result this time

- Corrected test: OD pairs are now sampled from 176 nodes within ~2,000 ft of the candidate, not citywide.
- These local trips plausibly used the interior streets before closure — and the result is still 0.0%.
- Manhattan's dense grid has enough equal-length parallel routes that one small closure rarely lengthens any trip.
- The perimeter number at left (57,014 → 58,258) isn't a second measurement — it's arithmetic: existing perimeter load plus 100% of the closed interior volume, an upper bound assumption.
- Caveat: distance only — turn penalties and signal delay aren't captured, so real driving-time cost is likely understated.

Manhattan's constraint isn't traffic volume — it's the density of bus and truck routes threading through a much finer grid than Barcelona's.

23.8% of the grid structurally qualifies — concentrated in upper Manhattan (Washington Heights, Marble Hill, Fort George), where side streets carry fewer bus and truck routes than the dense core — and that ranking is stable across every weighting scheme we tested. Barcelona's program targets traffic volume directly; on this data, that's the criterion doing the least work in Manhattan.

A ranked shortlist of 1,483 structurally-feasible Manhattan superblock candidates, located and named — and a quantified account of what rules the rest out.

Feasibility screening, not a design proposal. Community, political, and economic factors — including gentrification risk documented in Barcelona — sit outside the data. Traffic-volume screening is under-powered (142/8,208 edges measured), and the disruption test has been corrected to sample local trips, not just citywide ones — both discussed as limitations, not hidden.

Key reference: Eggimann, S. (2022). The potential of implementing superblocks for multifunctional street use in cities. *Nature Sustainability*, 5, 406–414.

Tools: osmnx, networkx, geopandas, shapely, pandas, numpy, requests, matplotlib, scipy

